

Phlebotomine sandflies (Diptera: Psychodidae) of southern Morocco: results of entomological surveys along the Marrakech–Ouarzazat and Marrakech–Azilal roads

S. BOUSSAA^{*,†}, M. NEFFA^{*}, B. PESSON[†] and A. BOUMEZZOUGH^{*}

^{*}*Equipe d'Ecologie Animale et Environnement, Faculté des Sciences Semlalia, Université Cadi Ayyad, Marrakech, Morocco*

[†]*Laboratoire de Parasitologie (JE 2533), Faculté de Pharmacie, Université de Strasbourg, B.P. 60024, 74 Route de Rhin, 67401 Illkirch, France*

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Since the 1970s, Azilal and Ouarzazat have been the main foci for human cutaneous leishmaniasis (CL) in Morocco. The sandflies along the main roads linking these two foci to Marrakech city, which is considered to be an area at risk of CL, were recently surveyed. Among the 872 sandflies collected, in June 2005, on the Marrakech–Ouarzazat road, *Sergentomyia fallax* was the most common species (36.1%), followed by *Phlebotomus sergenti* (21.1%), *P. papatasi* (14.2%), *S. minuta* (11.7%), *P. longicuspis* (5.5%), *P. alexandri* (5.4%), *P. perniciosus* (4.1%), *P. ariasi* (0.9%), *S. africana* (0.6%) and *S. dreyfussi* (0.3%). On the Marrakech–Azilal road, however, *S. minuta* was by far the most prevalent species (63.5% of the 1983 sandflies that were collected in August 2006), followed by *S. fallax* (12.9%), *P. perniciosus* (12.4%), *P. sergenti* (4.0%), *P. longicuspis* (3.0%), *P. papatasi* (2.8%), *S. dreyfussi* (1.1%) and *P. alexandri* (0.2%). The distribution of potential vectors along the two transects, according to altitude and bioclimate, was explored.

Human cutaneous leishmaniasis (CL) is a serious public-health problem in Morocco, especially in the two main foci, in the provinces of Azilal and Ouarzazat, that have been active since the 1970s. Ouarzazat is a focus of zoonotic CL caused by *Leishmania major*, with *Meriones shawi* acting as the main reservoir host and *Phlebotomus papatasi* as the vector (Rioux *et al.*, 1986). Azilal, in contrast, is a focus of anthroponotic CL caused by *L. tropica*, with *P. sergenti* identified as the main vector (Pratlong *et al.*, 1991).

In 2007, according to the Moroccan Ministry of Health (Anon., 2008) and

despite a national programme against leishmaniasis, 259 cases of CL were reported in Ouarzazat city and 365 in Azilal town. Although data collected since 1997 indicate that the incidence of CL in Ouarzazat city peaked in 2003, and has since fallen slightly, the incidence of the disease in the Azilal focus continues to increase (Fig. 1).

Both *P. papatasi* and *P. sergenti* have been collected and studied in the central–southern city of Marrakech (Boussaa *et al.*, 2005, 2007). Although the establishment of *P. papatasi* in the city appears to have been favored by urbanization, *P. sergenti* is rare in intensely urban sites and only relatively common in semi-rural settings (Boussaa *et al.*, 2007). The existence of populations of both potential vectors has raised concerns that Marrakech may become a new focus of CL, especially since, each year, large numbers of people migrate to the city from the

Reprint requests to: S. Boussaa, Laboratoire de Parasitologie (JE 2533), Faculté de Pharmacie, Université de Strasbourg, B.P. 60024, 74 Route de Rhin, 67401 Illkirch, France.

E-mail: samia.boussaa@pharma.u-strasbg.fr.

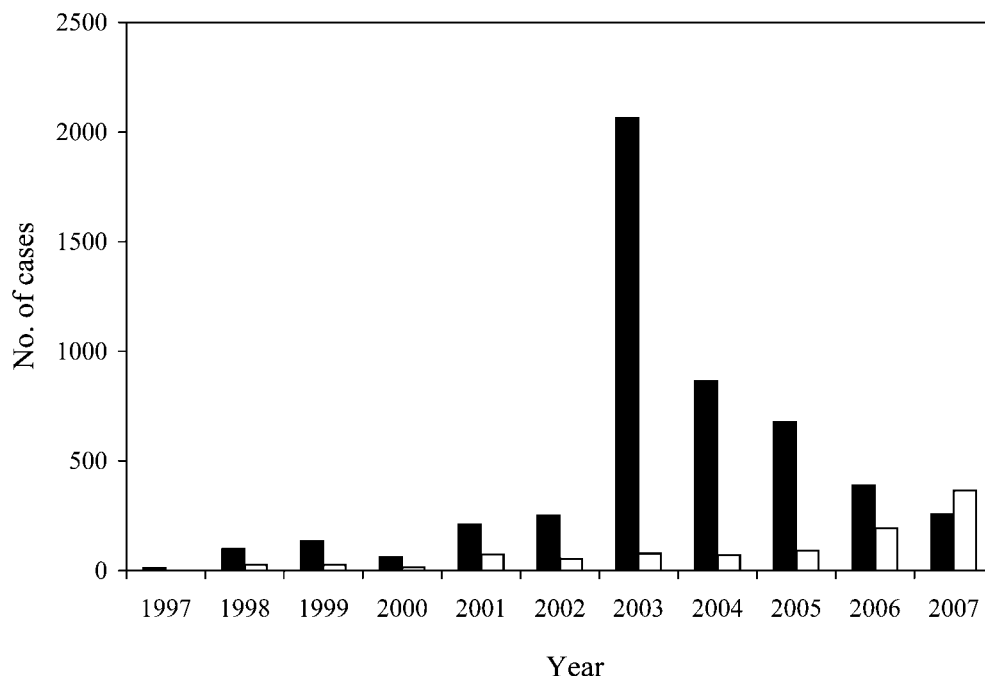


FIG. 1. The annual incidence of human cutaneous leishmaniasis, as reported in the Ouarzazat (■) and Azilal (□) areas between 1997 and 2007 (Anon., 2008).

Ouarzazat and Azilal provinces. The main aim of the present study was to explore the sandfly species present on the routes followed by these migrants, on the main roads linking Marrakech with the foci in Ouarzazat and Azilal.

MATERIALS AND METHODS

Transects

The first transect was 204 km long (Table 1 and Figure 2) and linked the cities of Marrakech and Ouarzazat. Marrakech lies on the semi-arid Al Haouz plain. Ouarzazat city, in central-southern Morocco, faces the desert at the foot of the High Atlas mountains and has an arid climate. The transect went through the Tizi-n-Tichka pass (at 2260 m above sea level) as well as eight collection sites on the northern slopes of the High Atlas, with either semi-arid (Tafriat, Touama, and Douar Igmir) or sub-humid, mountainous climates (Toufliht,

Source, Had Zerkten, Arghal, and Tadart) and another eight sites on the southern slopes, in either semi-arid (Aguelmous, Ighram, Aguim, and Tagouimalt) or arid (IminTiflet, Amezgan, Tabourihit, and Fedragon) areas.

The second transect (Table 2 and Figure 2) was 242 km long and connected Marrakech city to the town of Azilal. Azilal is in central Morocco, on the edge of the Mid Atlas mountains, and has a sub-humid climate. The collection sites along this transect, beyond Marrakech and Azilal, were located either on the semi-arid plain (Zemrane, Oued Lakhder, Ait Majden, and Tanant) or on the sub-humid northern foothills of the Atlas mountains (Abadou, Lahrouna, Imin'Ifri, and Damnate).

Sandfly Collection and Identification

Sandflies were collected using sticky paper traps (each an A4 sheet of paper coated with castor oil), on the nights of 12 June 2005 (Marrakech–Ouarzazat) or 16 August 2006

TABLE 1. Numbers of male (M) and female (F) sandflies collected along the Marrakech–Ouarzazat transect on the night of 12 June 2005

No. of males/no. of females collected														
Code	Site	Co-ordinates	Altitude (m)	Phlebotomus					Sergentomyia				No. of sandflies collected	
				papatasi	sergenti	alexandri	perniciosus	longicuspis	ariasi	fallax	minuta	africana		dreyfussi
S0	Marrakech city	31°36'N, 8°02'W	471	25/19	7/9	0/0	0/0	11/2	0/0	20/15	14/22	0/0	0/0	144
1	Tafiat	31°32'N, 7°36'W	755	3/0	4/2	0/0	0/0	0/0	0/0	0/1	1/0	0/0	0/0	11
2	Touama	31°31'N, 7°28'W	1088	1/1	16/2	0/0	4/0	3/0	0/0	0/0	4/4	0/0	1/0	36
3	Douar Igmir	30°21'N, 7°96'W	1753	5/1	24/6	0/0	0/0	0/0	0/0	1/0	1/2	0/0	0/0	40
4	Toufihit	31°29'N, 7°27'W	1400	0/0	0/0	0/0	1/0	0/0	1/0	0/0	0/0	0/0	0/0	2
5	Toufihit Source	31°29'N, 7°27'W	1459	0/0	0/0	0/0	0/0	0/0	1/0	0/0	0/0	0/0	0/0	1
6	Had Zerkten	31°25'N, 7°24'W	1360	0/0	1/0	0/0	0/0	0/0	0/0	0/0	0/2	0/0	0/0	3
7	Arghal	30°41'N, 6°37'W	1439	0/0	10/4	0/0	12/2	1/0	0/1	0/0	1/0	0/0	0/0	31
8	Tadart	30°58'N, 6°42'W	1650	0/0	2/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	2
9	Aguelmous	31°15' N, 7°24'W	2123	0/0	1/1	0/0	6/0	0/0	3/1	2/1	1/0	0/0	0/0	16
10	Ighram	31°07'N, 7°25'W	1995	0/0	0/0	0/0	3/0	1/0	1/0	0/0	0/1	0/0	0/0	6
11	Aguin	31°09'N, 7°28'W	1790	4/1	8/3	0/0	4/1	1/0	0/0	1/0	0/0	0/0	0/0	23
12	Tagouimat	30°37'N, 7°34'W	1866	3/1	2/0	0/0	3/0	12/4 + 1gr	0/0	0/0	0/1	0/0	0/0	27
13	Imin Tiflet	31°06'N, 7°16'W	1495	7/0	5/1	0/0	0/0	0/0	0/0	0/0	0/1	0/0	0/0	14
14	Amezgan	31°02'N, 7°12'W	1309	13/7	42/10 + 3gr	0/1 + 1gr	0/0	8/0	0/0	3/3	2/1	4/1	1/0	100
15	Tabourihit	30°58'N, 7°07'W	1258	9/4	12/2	34/11	0/0	0/0	0/0	136/128	34/10	0/0	0/0	380
16	Fedragon	30°55'N, 6°58'W	1132	14/4	6/1	0/0	0/0	3/1	0/0	1/1	0/0	0/0	0/0	31
17	Ouarzazate city	30°55'N, 6°55'W	1151	1/1	0/0	0/0	0/0	0/0	0/0	1/1	0/0	0/0	0/1	5
–	All 18	–	–	85/39	140/44	34/13	33/3	40/8	6/2	165/150	58/44	4/1	2/1	872

gr, Gravid female.

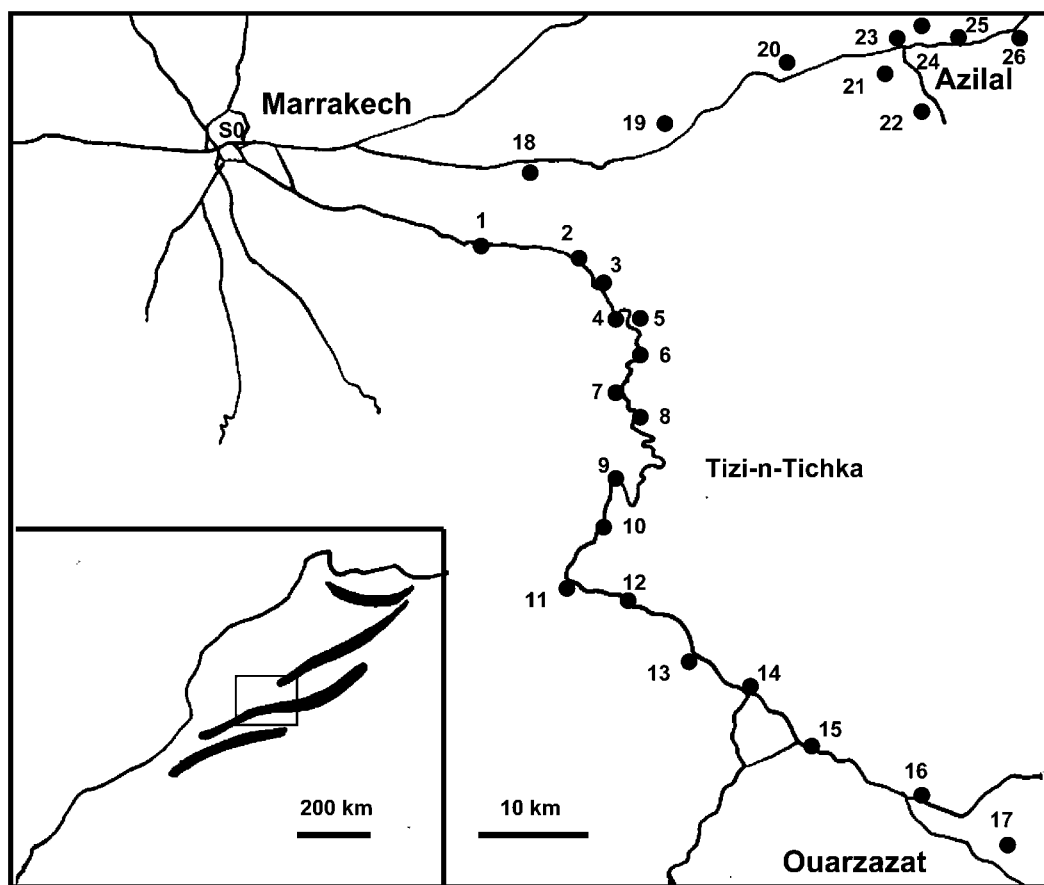


FIG. 2. Sketch map of study area, showing the locations of the 27 sandfly-collection sites.

(Marrakech–Azilal). Two collections were made at the site (Akioud, in Marrakech city) that lay on both transects.

Specimens were preserved in 70% ethanol, cleared in 20% (w/v) KOH and Marc-André solution, and then mounted on microscope slides, in Hoyer's medium. Most of the sandflies were then identified to species by the morphological examination of the genitalia (males) or the pharyngeal armature and spermathecae (females), with the help of the keys and descriptions published by the Moroccan Ministry of Health (Anon., 1997). For the *P. perniciosus* complex, however, the females were identified by examining the dilatation of the distal parts of the spermathecal ducts (Léger *et al.*, 1983) and males by examining the shape of

the copulatory valves and counting the number of coxite hairs (Benabdennbi *et al.*, 1999; Boussaa *et al.*, 2008a).

RESULTS AND DISCUSSION

Overall, 872 sandflies were collected along the Marrakech–Ouarzazat transect (Table 1) and 1983 on the Marrakech–Azilal transect (Table 2). In the combined collections, six *Phlebotomus* species (*P. papatasi*, *P. sergenti*, *P. alexandri*, *P. longicuspis*, *P. perniciosus* and *P. ariasi*) and four *Sergentomyia* species (*S. minuta*, *S. fallax*, *S. dreyfussi* and *S. africana*) were detected.

On the Marrakech–Azilal transect, *S. minuta* was the most prevalent species

TABLE 2. Numbers of male (M) and female (F) sandflies collected along the Marrakech–Azilal transect on the night of 16 August 2006

Code	Site	Co-ordinates	Altitude (m)	No. of males/no. of females collected										No. of sandflies collected
				Phlebotomus					Sergentomyia					
				papatasi	sergenti	alexandri	pernicius	longicuspis	fallax	minuta	dreyfussi			
S0	Marrakech city	31°36'N, 8°02'W	471	16/7	6/1	0/0	0/0	3/0	19/12	4/3	0/0	71		
18	Zemrane	31°53'N, 8°26'W	424	6/10	0/0	0/0	7/0	3/0	1/0	14/45	0/5	91		
19	Abadou	31°26'N, 7°31'W	1638	0/1gr	0/0	1/0	23/1	0/2	22/14	200/175	2/7	448		
20	Lahrouna	31°07'N, 7°28'W	1260	0/0	7/0	0/0	108/2	11/0	15/23	80/118	2/6	372		
21	Imi n'Ifri	30°03'N, 8°40'W	1037	0/0	29/0	0/0	42/1	1/0	2/4	78/96	0/0	253		
22	Damnate	31°73'N, 7°00'W	968	13/1	19/2 + 1eng	0/0	21/1	27/1 + 1eng	35/38 + 5gr + 1eng	150/136 + 3gr + 2eng	0/0	457		
23	Oued Lakhder	32°49'N, 8°14'W	145	0/0	0/0	2/0	5/0	4/1	0/0	1/0	0/0	13		
24	Ait Majden	31°84'N, 6°96'W	705	2/0	4/0	0/0	23/1gr	3/0	5/8	23/17 + 1eng	0/0	87		
25	Tanant	31°86'N, 6°95'W	889	0/0	2/5	0/0	9/1	3/0	11/9	30/36	0/0	106		
26	Azilal	31°58'N, 6°34'W	1351	0/0	3/1	0/0	1/0	0/0	14/16 + 2gr	26/21 + 1gr	0/0	85		
—	All 10	—	—	37/19	70/10	3/0	239/7	55/5	124/132	606/654	4/18	1983		

eng, Engorged female; gr, gravid female.

(63.5%), followed by *S. fallax* (12.9%), *P. perniciosus* (12.4%), *P. sergenti* (4.0%), *P. longicuspis* (3.0%), *P. papatasi* (2.8%), *S. dreyfussi* (1.1%) and *P. alexandri* (0.2%). Although the majority of the collected sandflies belonging to the genus *Phlebotomus* were male, more female than male *Sergentomyia* were caught.

Along the Marrakech–Ouarzazat transect, *S. fallax* was the most prevalent species (36.1%; particularly common in Tabourihit), followed by *P. sergenti* (21.1%), *P. papatasi* (14.2%), *S. minuta* (11.7%), *P. longicuspis* (5.5%), *P. alexandri* (5.4%), *P. perniciosus* (4.1%), *P. ariasi* (0.9%), *S. africana* (0.6%) and *S. dreyfussi* (0.3%). For each sandfly species, the males collected outnumbered the females. Only 270 of the sandflies caught along this transect came from the northern slopes of the High Atlas (i.e. between Marrakech and Tichka), the other 602 specimens being collected on the southern slopes (i.e. between Tichka and Ouarzazat). In the collections made between Marrakech and Tichka, *P. sergenti* was the predominant species (32.2%), followed by *P. papatasi* (20.4%), *S. minuta* (18.9%), *S. fallax* (13.7%), *P. perniciosus* (7.0%), *P. longicuspis* (6.3%), *P. ariasi* (1.1%) and *S. dreyfussi* (0.4%), with *P. alexandri* and *S. africana* both absent. Between Tichka and Ouarzazat, however, *S. fallax* was the predominant species (46.2%, collected mainly in one locality), followed by *P. sergenti* (16.1%), *P. papatasi* (11.5%), *S. minuta* (8.5%), *P. alexandri* (7.8%), *P. longicuspis* (5.2%), *P. perniciosus* (2.8%), *P. ariasi* (0.8%), *S. africana* (0.8%) and *S. dreyfussi* (0.3%).

No morphological anomalies were observed in any of the female sandflies that were examined but every male *P. perniciosus* collected had a single tipped copulatory valve that was curved at its apex. This 'atypical' form of male *P. perniciosus* appears to be the only form found in southern Morocco (Benabdennbi *et al.*, 1999; Pesson *et al.*, 2004; Boussaa *et al.*, 2008a).

In addition, a few of the male *P. papatasi* collected in Marrakech city had an extra spine or just one spine on each of their surstyles (Boussaa *et al.*, 2008b).

Over the last two decades, foci of anthroponotic CL, caused by *L. tropica*, have been identified or confirmed to be active in many areas of Morocco: in Azilal, Essaouira (Pratlong *et al.*, 1991), Taza (Guessous-Idrissi *et al.*, 1997; Guessous-Idrissi, 1999), Zouagha Moulay Yaacoub (Rhajaoui *et al.*, 2004), Chichaoua (Guernaoui *et al.*, 2005) and Al Haouz (Ramaoui *et al.*, 2008; Boussaa *et al.*, 2009). In addition, in southern Morocco (in Ouarzazat, Errachidia and Zagora), zoonotic CL, caused by *L. major*, remains a major public-health problem (Anon., 2008).

In Morocco, *P. papatasi* and *P. sergenti* are known to be primary vectors, of *L. major* and *L. tropica*, respectively (Rioux *et al.*, 1986; Pratlong *et al.*, 1991). Although *P. perniciosus*, the proven vector of *L. infantum* in the Mediterranean basin (Killick-Kendrick, 1990), is not uncommon in Morocco, the epidemiological role, if any, of the atypical form seen in the present study — the only morphotype found in southern Morocco — remains to be elucidated (Pesson *et al.*, 2004; Boussaa *et al.*, 2008a).

The present entomological investigations were carried out in the northern foothills of the High Atlas mountains and on the plain of Al Haouz (Marrakech–Azilal transect) and on both the northern and southern slopes of the mountains (Marrakech–Ouarzazat transect), at sites with various bioclimates. In total, over just one night at each collection site, 2855 sandflies representing 10 species were collected. Along the Marrakech–Ouarzazat transect, a vector, *P. sergenti* (21.1%) appeared to be the most common *Phlebotomus* species, followed by another vector, *P. papatasi* (14.2%). Most of the *P. sergenti* collected came from the Ouarzazat area (Amezgan and Tabourihit), whereas most of the *P. papatasi* came from the other end of the transect, in and around the Marrakech area. On the

Marrakech–Azilal transect, the most common *Phlebotomus* species were again known vectors of leishmanial parasites — *P. perniciosus* (12.4%) and *P. sergenti* (4.3%) — with most specimens of these two species being caught in the Azilal area (in Ait Majden, Tanant and Azilal city).

When the collection data on the potential vectors were explored in terms of the altitude of each collection site, it became apparent that *P. papatasi* was most abundant at sites lying at 400–800 m above sea level whereas *P. perniciosus* and *P. sergenti* were most common in the collections made 1200–1600 m above sea level on the Marrakech–Ouarzazat transect and 800–1200 m above sea level on the (generally lower) Marrakech–Azilal transect. In south-western Morocco, Guernaoui *et al.* (2006) collected *P. sergenti* at all altitudes and found the correlations between altitude and the abundance of *P. sergenti*, *P. perniciosus* and *P. papatasi* to be slightly positive, highly positive and slightly negative, respectively.

The results of several previous studies in Morocco (Rioux *et al.*, 1984, 1997; Rioux, 2001; Rispaïl *et al.*, 2002; Rioux and de La Rocque, 2003) have indicated that the distribution of sandflies is mainly determined by bioclimate, with the risk zones for CL being defined by the bioclimatic ‘preferences’ of each vector: per-arid and arid for *P. papatasi*, semi-arid for *P. sergenti*, and humid or sub-humid for members of the subgenus *Larrousius*, including *P. perniciosus* (Rioux *et al.*, 1984, 1997). Although such ‘preferences’ may explain why, in the present study, *P. papatasi* was not found at the collection sites in sub-humid areas, they do not explain why *P. sergenti* was present in all the sampled bioclimates or why *P. perniciosus* tended to be most common in arid areas along the transects. Boussaa *et al.* (2008a) suggested that, compared with the typical morphotype, the atypical morph of *P. perniciosus* found in southern Morocco could be more adapted to the aridity of this area.

The presence of *P. sergenti* at many altitudes and in all the bioclimates sampled

along both transects may help explain why anthroponotic CL is relatively widespread in Morocco, although it remains to be seen whether all of the Moroccan sandflies currently assigned to *P. sergenti* have similar epidemiological importance. The results of the recent investigation of mitochondrial-DNA haplotypes (Yahia *et al.*, 2004; Barón *et al.*, 2008) and isoenzyme analysis (Boussaa *et al.*, 2009) indicate that the taxon known as *P. sergenti* in Morocco does not have the genetic characteristics of a single species. The distinct lineages or cryptic species present in the country may well differ in their vectorial capacities and mechanisms of parasite transmission (Depaquit *et al.*, 2002; Yahia *et al.*, 2004).

In Morocco, before risk zones for CL can be accurately defined and an efficient control strategy can be established, the epidemiological significance of the genetic variability in ‘*P. sergenti*’ and of the presence of an atypical morph of *P. perniciosus* need to be explored.

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